HOPSTOP — OVERALL SYSTEM ARCHITECTURE & PRE-SETTLEMENT DATAFLOW ON IBM Z / LINUXONE

Track 1: Real-Time AI for Critical Decisions • IBM Z Datathon 2026 • Author: Kabir Roy (VIT Bhopal)

1. PAYMENT STREAM INGRESS

Instant Settlement Rails

Chronological Event Replay

UPI / IMPS / ACH / Wire

50,880 AMLworld Transactions

5,000 Accounts • 24 Banks

8 Laundering Typologies

- Fan-Out & Fan-In Bursts
- Gather-Scatter & Scatter-Gather
- Simple-Cycle, Stack, Bipartite

Benign Hard-Negative Control

Corporate Payroll & Vendor Batches

(High fan-out, 0.0 pass-through)

Raw Event Payload Schema

```
tx_id: 35861
ts: 2026/09/08 05:11
from_bank: 14 → to_bank: 27
amount: INR 185,000.00
payment_format: "Wire"
```

Zero Network Hop (In-Memory)

2. IBM Z / LINUXONE (s390x) ON-PLATFORM CO-LOCATED INFERENCE & CRYPTO ENCLAVE

SLA < 10 ms • A

STAGE 2A • CPACF CRYPTO (FACILITY 158)

Zero-PII Account Hashing

Salted SHA-256 Before Feature State

```
SHA256(salt || raw_acct_id)
→ ACCT_EF72606626ED
```

Raw account IDs never enter ML

- Hardware-assisted on s390x CPACF
- Prevents PII exposure across logs
- Deterministic cross-bank entity key

STAGE 2B • CAUSAL O(1) FEATURE ENGINE

12 Graph-Velocity Features

Bounded 15m / 1h / 24h Deques (Past Events Only)

1. pass_through_ratio_15m (0.96x)
2. rapid_turnover_flag (1.0)
3. fan_in_1h / fan_out_1h (O(1) deque)
4. fan_in_24h / fan_out_24h
5. counterparty_novelty (new vs known)
6. amount_zscore (Welford O(1))
7. time_since_credit_log & format_risk

STAGE 2C • IBM SNAP

Tensor-Tree GBD

tree_format='auto'

Hardware Runtime Pr

- /proc/cpuinfo Faci
- libzdnn: zdnn_is_
- Matrix Tensor-Tree

Latency: p50 0.040 ms |

Throughput: 109,612 TP

STAGE 2D • ALERT-BUDGET POLICY CALIBRATOR & SHAP EXPLAINABILITY (hopstop/pol

Calibrated 3-Tier Policy (0.5%–2.5% Alert Budget) + Top-3 SHAP Reason

HOLD (p ≥ 0.9940)

Freeze transfer pre-settlement

Severs Hop 1 / Hop 2 mule chain

STEP_UP (p ≥ 0.9896)

Biometric / OTP + 15m Cool-Off

Stops APP scam social engineering

ALLOW (

Instant sett

Payroll safe

Top-3 SHAP: [1] Rapid Drain (+4.70) [2] Fan-Out (+3.25) [3] Wire Velocity (+3.18)

Payroll Differentiator (fan_out=5, pass_through=0.0): Rules → FLAGGED | HopStop → ALLOW (0.008

STAGE 2E • TAMPER-EVIDENT LEDGER

SHA-256 Hash-Chain Audit

1930 Cybercrime & FIU-IND Ready

```
H_n = SHA256(H_{n-1} || tx_id
|| score || action || reasons)
sig: 05cd88d9553492d8f9a17c...
```

- Immutable regulatory evidence
- Instant 1930 helpline lien packet

4. PRE-SETTLEMENT MULTI-HOP LAUNDERING CHAIN INTERDICTION TOPOLOGY (94.0% OF TEST CHAINS FROZEN AT HOP 1 OR HOP 2)

By holding the outgoing transfer at Hop 1 or Hop 2 before settlement completes, HopStop prevents stolen victim funds from dispersing across downstream banks:

ORIGIN • VICTIM ACCOUNT

APP Scam / Phishing Drain

ACCT_EF72606626ED (Bank #14)

Initial compromised transfer

Inflow: INR 192,000

+4m drain

HOP 1 • MULE DISPERSAL [HOLD • p=0.994]

Pre-Settlement Freeze in 0.05 ms

ACCT_EF7260 → ACCT_AB5796 (Bank #27)

pass_through_15m = 0.96 • fan_out_1h = 5

INR 185,000 HELD ON-PLATFORM

✕ SEVERED

HOP 2 • DOWNSTREAM MULE [BLOCKED_UPSTREAM]

Layering Fan-Out Prevented

ACCT_AB5796 → 3 Cross-Bank Accounts

Funds never reach Hop-2 mule accounts

INR 185,000 Protected for Victim Lien